D-ANMOL IT SOLUTIONS

Generative AI

Use generative AI tools and concepts to create content, research, and intelligent applications.

6 Months | Advanced Professional Track

A step-by-step practical learning roadmap with explained tools, applied examples, project milestones and career preparation.

01 WHAT CAN YOU BUILD AFTER THIS COURSE?

- AI Content Workflow: Create a repeatable content research, drafting and review workflow.
- Knowledge Base Q&A Prototype: Answer questions from an approved set of documents with source references.

Possible career directions

GenAI application trainee, AI content specialist, prompt workflow assistant and AI automation trainee. Roles are illustrative and depend on skills, experience and hiring criteria.

Your learning journey

From learning to practical delivery

Learn concepts	**Practise skills**	**Build projects**	**Show portfolio**

02 STEP-BY-STEP COURSE SYLLABUS

Modules are ordered from foundations to application. Each includes what you learn, why it matters and a practical classroom activity.

STEP 01 | Generative AI Foundations

What you will study:

- AI vs ML vs GenAI: understand the concept, practise it with a guided example, and apply it in a relevant task.
- generative models: understand the concept, practise it with a guided example, and apply it in a relevant task.
- LLM concepts: understand the concept, practise it with a guided example, and apply it in a relevant task.
- tokens: understand the concept, practise it with a guided example, and apply it in a relevant task.
- context windows and use cases: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Identify suitable and unsuitable tasks for generative AI.

STEP 02 | Prompt Engineering Basics

What you will study:

• Role: understand the concept, practise it with a guided example, and apply it in a relevant task.

• task: understand the concept, practise it with a guided example, and apply it in a relevant task.

• context: understand the concept, practise it with a guided example, and apply it in a relevant task.

• constraints: understand the concept, practise it with a guided example, and apply it in a relevant task.

• examples: understand the concept, practise it with a guided example, and apply it in a relevant task.

• output formats and iterative prompting: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Create reusable prompts for common work tasks. D-ANMOL IT SOLUTIONS | COURSE CURRICULUM Page 2

STEP 03 | Research & Information Workflows

What you will study:

• Research planning: understand the concept, practise it with a guided example, and apply it in a relevant task.

• source checking: understand the concept, practise it with a guided example, and apply it in a relevant task.

• summarisation: understand the concept, practise it with a guided example, and apply it in a relevant task.

• comparison and hallucination awareness: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: It provides a foundation for later modules and real project implementation.

Hands-on example: Prepare a source-checked research brief with AI assistance.

STEP 04 | Writing & Communication

What you will study:

• Emails: understand the concept, practise it with a guided example, and apply it in a relevant task.

• reports: understand the concept, practise it with a guided example, and apply it in a relevant task.

• proposals: understand the concept, practise it with a guided example, and apply it in a relevant task.

• social captions: understand the concept, practise it with a guided example, and apply it in a relevant task.

• tone control and editing workflows: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create a consistent multi-format content set.

STEP 05 | Image, Design & Presentation AI

What you will study:

• Image prompting: understand the concept, practise it with a guided example, and apply it in a relevant task.

• visual direction: understand the concept, practise it with a guided example, and apply it in a relevant task.

• design iteration: understand the concept, practise it with a guided example, and apply it in a relevant task.

• slide outlines and asset review: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create a campaign visual concept and presentation.

STEP 06 | Productivity & Data Assistance

What you will study:

• Spreadsheet formulas: understand the concept, practise it with a guided example, and apply it in a relevant task.

• data explanation: understand the concept, practise it with a guided example, and apply it in a relevant task.

• document extraction and meeting summaries: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a repeatable productivity workflow.

STEP 07 | LLM Concepts & API Introduction

What you will study:

• How LLM APIs work: understand the concept, practise it with a guided example, and apply it in a relevant task.

• API keys: understand the concept, practise it with a guided example, and apply it in a relevant task.

• requests: understand the concept, practise it with a guided example, and apply it in a relevant task.

• responses: understand the concept, practise it with a guided example, and apply it in a relevant task.

• tokens: understand the concept, practise it with a guided example, and apply it in a relevant task.

• latency and cost awareness: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Call an available LLM API from a simple test application.

STEP 08 | Python for AI Applications

What you will study:

• Python essentials: understand the concept, practise it with a guided example, and apply it in a relevant task.

• JSON: understand the concept, practise it with a guided example, and apply it in a relevant task.

• HTTP requests: understand the concept, practise it with a guided example, and apply it in a relevant task.

• environment variables and error handling: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create a small command-line AI assistant.

STEP 09 | Embeddings & Semantic Search

What you will study:

• Embeddings: understand the concept, practise it with a guided example, and apply it in a relevant task.

• similarity: understand the concept, practise it with a guided example, and apply it in a relevant task.

• chunking: understand the concept, practise it with a guided example, and apply it in a relevant task.

• vector search and retrieval concepts: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Build a small semantic search prototype.

STEP 10 | RAG & Knowledge Assistants

What you will study:

• Document ingestion: understand the concept, practise it with a guided example, and apply it in a relevant task.

• retrieval: understand the concept, practise it with a guided example, and apply it in a relevant task.

• grounding: understand the concept, practise it with a guided example, and apply it in a relevant task.

• citations: understand the concept, practise it with a guided example, and apply it in a relevant task.

• evaluation and failure cases: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Create a document Q&A proof of concept.

STEP 11 | AI Agents & Multimodal Workflows

What you will study:

• Tools: understand the concept, practise it with a guided example, and apply it in a relevant task.

• agent loops: understand the concept, practise it with a guided example, and apply it in a relevant task.

• structured outputs: understand the concept, practise it with a guided example, and apply it in a relevant task.

• image/audio use cases and human review: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Prototype a controlled multi-step assistant. D-ANMOL IT SOLUTIONS | COURSE CURRICULUM Page 3

STEP 12 | Capstone, Safety & Deployment Basics

What you will study:

- Privacy: understand the concept, practise it with a guided example, and apply it in a relevant task.
- copyright: understand the concept, practise it with a guided example, and apply it in a relevant task.
- evaluation: understand the concept, practise it with a guided example, and apply it in a relevant task.
- prompt injection awareness: understand the concept, practise it with a guided example, and apply it in a relevant task.
- documentation and demo: understand the concept, practise it with a guided example, and apply it in a relevant task.

Why this matters: You use this skill to solve practical problems, improve accuracy and build reliable outputs.

Hands-on example: Present a useful GenAI solution with limitations documented.

03 TOOLS, LIBRARIES & THEIR USES

This section explains where each tool or library fits in a practical project. Some advanced tools may be introduced as demonstrations in shorter tracks.

Tool / library	What it is used for
LLM tools	Generate, revise and summarize text with human verification.
Python	Connect workflows, handle data and build prototypes.
Embeddings	Represent text as vectors for semantic search.
Vector database / FAISS	Retrieve relevant document passages for question answering.
LLM API	Connect a model to an application with secure credentials.

04 PROJECT-BASED LEARNING

Included: 2 project(s). Every project includes planning, implementation, testing and a presentation of results.

LIVE PROJECT 1: Knowledge Base Q&A Prototype

What you will build: Answer questions from an approved set of documents with source references.

Where tools are used: Python, document chunking, embeddings, vector search and an LLM.

Final deliverable: Q&A prototype with evaluation examples.

Project workflow

Input / brief	Build & test	Review results	Present output

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

LIVE PROJECT 2: AI Marketing Assistant

What you will build: Generate campaign briefs and draft variants under human approval.

Where tools are used: Structured prompts, Python JSON handling and optional LLM API.

Final deliverable: Prototype, output samples and safety controls.

Project workflow

Input / brief	Build & test	Review results	Present output

Live project means an applied, end-to-end project based on a realistic business scenario or suitable dataset; access to an external client or production deployment is not implied.

05 TRAINING PLAN & ASSESSMENTS

Suggested duration: 6 Months (approximately 24 weeks). Exact session hours and timetable should be confirmed with the institute.

- Month 1–2 — Foundations, tools and regular practice
- Month 3–4 — Intermediate/advanced topics and first live project
- Month 5 — Second live project, documentation and portfolio
- Month 6 — Interview classes, mock interviews and capstone presentation

Evaluation: practical exercises, module checkpoints, code/document review where applicable, project demonstration and final feedback.

06 INCLUDED INTERVIEW CLASSES

- Resume and portfolio presentation for the selected course.
- Technical and practical interview questions based on course modules.
- Mock interviews, communication practice and feedback.
- Explaining project architecture, decisions, challenges and results.

FREE SUPPORT INCLUDED WITH EVERY COURSE

1 YEAR DOUBT SUPPORT + INTERVIEW PREPARATION — FREE OF COST

- One year of doubt-clearing support related to the enrolled course.
- Interview preparation support at no additional course fee.
- Guidance on explaining course projects and strengthening fundamental concepts.
- Availability, format and scheduling of support sessions to be communicated by the institute.

Important: This brochure presents the requested training offer. The institute should confirm the exact support policy, availability, start date and any applicable conditions before publishing. Interview preparation is not a job or placement guarantee.

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